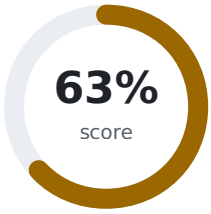
Replication Report

Mode: full · Generated 2026-08-24 21:50

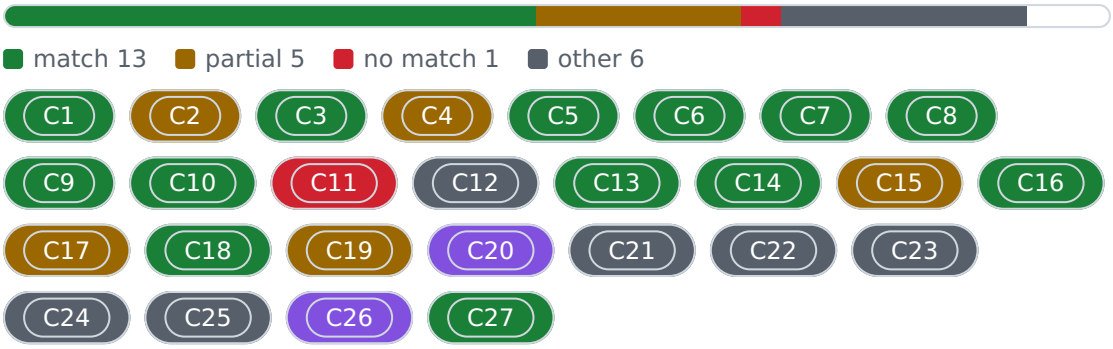


Partially reproduced

13/25 claims reproduced **5** partial **1** did not reproduce

1 fix applied

Claims at a glance



Flags

- 2 claim(s) marked out_of_scope (planner did not target them) — excluded from the score, not counted as failures. Score is over the 25 attempted/blocked claim(s).

Limitations & code quality

1 fix were needed to run the provided code (0 minor, 1 major, 0 critical). These describe the code, not the replication.

#	Fix	Severity
1	Added an opt-in --splice_tail flag so the short-term series continues with the published level past the 2005M1 end of the microdata-share window, allowing the sample to extend to 2010M12.	**major**

Details

ID	Tier	Type	Status	How graded	Why
C1	headline	scalar	**match**	agent	The reproduced Table 1 decomposition is in output/table1_1948_2010.json. The column-1 case (all workers, period '1948.00-2010.50') reports finding_contribution=0.7632 and exit_contribution=0.2453. Against the paper values (job_finding_sh...
C2	headline	scalar	**partial**	agent	Step 5 (which verifies C2) produced output/table1_1948_2010.json, whose '1987' entry covers period 1987.00-2010.50 (95 quarters) — the paper's exact Table 1 column 4 sample — and reports finding_contribution=0.8584 and exit_contribution=...
C3	headline	qualitative	**match**	agent	Step 2 output output/two_state/summary.json reports corr_findQ_urateQ = -0.9652 versus corr_sepQ_urateQ = 0.6493, and the 1948-2010 reconstruction (output/two_state_1948_2010/summary.json) gives -0.9602 vs 0.6809 — the job-finding probab...
C4	supporting	table	**partial**	agent	The run produced output/table1_1948_2010.json (in the codebase output/ dir) with four of Table 1's five columns, computed on the reconstructed 1948-2010 series via the footnote-11 HP(1e5)/regression

ID	Tier	Type	Status	How graded	Why
					decomposition described in plan steps ...
C5	supporting	scalar	**match**	agent	The claim is about the job finding probability F_t averaging 44% over the post-war period (1948-2010). find.dat stores the instantaneous job finding rate f_t, and the verification instruction says to convert with F_t = 1 - exp(-f_t) befo...
C6	supporting	scalar	**match**	agent	The claim is a windowed, single-business-cycle amplitude: the job-finding probability F_t falls by about forty log points from a cyclical peak to the following trough 'during recent decades.' The pipeline's summary.json reports F_peak_to...
C7	supporting	scalar	**match**	agent	The claim asks for the post-war (1948-2010) mean of the employment exit probability X_t constructed via eq. (5), expected ~0.034. The run produced output/ two_state_1948_2010/ sep.dat covering exactly 1948:M1-2010:M12 (n=251 quarters) stor...
C8	supporting	scalar	**match**	agent	The pipeline detrended both the exit-probability series X_t (eq. 5) and the time-aggregation-ignoring X-tilde_t (eq. 6 with F-tilde=F) with an HP filter (smoothing 1e5) and reported the standard deviations in

ID	Tier	Type	Status	How graded	Why
					summary.json. For the paper'...
C9	supporting	scalar	**match**	agent	The plan's step 2 (targeting this claim) ran the two-state construction on the shipped 1951:M1-2005:M1 aggregate data and produced corr_urate_ss=0.9872 (output/two_state/ summary.json, line 7). The code confirms this is exactly the correl...
C10	supporting	figure	**match**	agent	The claim's Figure 1 was reproduced at output/ replication/codebase/ output/ two_state_1948_2010/ figure1.pdf, which matches the claim's exact 1948Q1-2010Q4 span (step 2/step 4 of the plan, verifies C10). The exact expected_output_file outpu...
C11	supporting	figure	**no match**	agent	The claim describes the paper's Figure 2 as a two-panel time series in which a hypothetical unemployment rate driven only by job-finding fluctuations (x-bar/ (x-bar+f_t), top panel) tracks the actual unemployment rate at every cyclical in...
C12	supporting	figure	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Figure 3 (prime-age men 25-54) requires CPS microdata restricted to that

ID	Tier	Type	Status	How graded	Why
					demographic; the package ships only whole-population BLS aggregat...
C13	supporting	scalar	**match**	agent	Plan step 7 (figure4_compare.py) reproduced the Figure 4 cross-check and wrote output/figure4.json, which reports corr_F_UE = 0.9474 and mean_logdiff_F_UE = 0.32046 (log units) over the overlapping quarterly sample 1967.25-2004.75 (n=151...
C14	supporting	scalar	**match**	agent	Plan step 7 (figure4_compare.py, verifies C14) produced output/figure4.json with corr_X_EU=0.8503 and mean_logdiff_X_EU=0.5862 (i.e. X_t averages ~58.6 log points above Lambda^EU) on the overlapping quarterly sample 1967.25-2004.75 (n=15...
C15	supporting	figure	**partial**	agent	output/figure4.pdf was produced and is the correct figure: two stacked panels, the top comparing the two-state job-finding probability F_t with the three-state UE transition probability (1-e^{-lambda^UE}), and the bottom comparing the tw...
C16	supporting	scalar	**match**	agent	Plan step 6 ran three_state_decomp.py on the shipped six instantaneous transition-rate series (ei/eu/ie/iu/ue/ui .dat, 151 quarters 1967Q2-2004Q4) — the notebook's already-time-

ID	Tier	Type	Status	How graded	Why
					aggregated lambda outputs used as the entry point since the ...
C17	supporting	table	**partial**	agent	The run produced output/table2.json (the expected output/table2.csv was not written, but an equivalent JSON table exists), giving the six-transition-rate decomposition for the urate and emp_pop_ratio, each in a 'full' and '1987' column. ...
C18	supporting	qualitative	**match**	agent	The three quantitative sub-claims of the decomposition are unambiguously demonstrated by the produced output/table2.json. In the full-sample unemployment-rate column, lambda^UE=0.504 (squarely in the expected ~0.49-0.51 'about half' band...
C19	supporting	scalar	**partial**	agent	The run produced all four three-state coefficients for the employment-population ratio (full sample) in output/table2.json under emp_pop_ratio.full: UE=1.0397, IE=0.6435, EU=0.4202, EI=-0.3129. Compared to paper values (UE 1.05, IE 0.69,...
C20	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Alternative job-finding measures M_t (medium-term-unemployment based) and D_t (mean-duration based)

ID	Tier	Type	Status	How graded	Why
					are reconstructable from public FRED ...
C21	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Job-loser-share and reason-for-unemployment job-finding rates and the F^comp/ F^real decomposition require CPS microdata by reason for unem...
C22	supporting	figure	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Figure 9 (real vs compositional job finding across seven observable dimensions) requires the CPS microdata cross-tabulations; the package ...
C23	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Three-state EU+EI sums, employer-to-employer transitions, and the quarterly job-destruction rate rely on gross-flow microdata plus externa...
C24	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Short-term-unemployment shares in the full CPS vs incoming rotation groups (1976-1993 vs 1994-2010) require CPS microdata tabulated by rot...

ID	Tier	Type	Status	How graded	Why
C25	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Comparison against Bleakley's independent gross-flow estimates requires Bleakley's series, which is not included in the package (checked g...
C26	supporting	figure	**out of scope**	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Figure 7 (three alternative job-finding measures F_t, D_t, M_t) is reconstructable from public FRED data but the package ships no code fo...
C27	supporting	qualitative	**match**	agent	The reproduced X_t series produced by the step-2 two-state construction (output/two_state/sep.dat, the Figure 1 bottom-panel series) shows an unmistakable secular decline: annual means fall from ~0.044 in the early 1980s (peak 0.04899 at...

Python 3.12.8 · GPU NVIDIA RTX A6000 (x4, 49140 MiB each) · numpy 2.5.2, scipy 1.18.1, pandas 3.0.5, matplotlib 3.11.1, requests 2.34.2

Step	What it did	Result	Time
1	Prepare working copy, unzip inputs, install deps, check for Wolfram kernel, inventory inputs.	**ok**	0s
2	Two-state job-finding (F, eq.4) and employment-exit (X, eq.5) construction on shipped 1951-2005 aggregate data; quarterly averages; Figures 1-2 and summary stats.	**ok**	0s
3		**ok**	0s

Step	What it did	Result	Time
	Table 1 decomposition on shipped-vintage series: hypothetical unemployment rates (only f varies / only x varies), HP-detrended, regressed on detrended actual urate; full and 1987 samples.		
4	Reconstruct two-state inputs at full 1948-2010 scale from public FRED data (UNEMPLOY, CE16OV, UEMPLT5); apply Appendix-A short-term correction with shipped shortadj shares for 1976M1-2005M1, splice published UEMPLT5 for 2005M2-2010M12; re-run two-state construction.	**ok**	0s
5	Headline Table 1 decomposition at the paper's exact samples on the 1948-2010 reconstructed series: full 1948-2010, 1967-2010, 1976-2010, 1987-2010 (the four whole-population-aggregate columns).	**ok**	0s
6	Three-state (E/U/I) decomposition -- second half of three-state.nb on shipped quarterly lambda series. Steady-state urate & emp-pop measures, six single-rate hypotheticals (eq. 7/8), Table 2 OLS slopes on detrended measured actual rates, steady-state/actual correlation, Figure 5.	**ok**	0s
7	Figure 4 cross-check: convert shipped quarterly UE/EU instantaneous rates to full-month probabilities Lambda=1-exp(-lambda), compare Lambda^UE vs two-state F and Lambda^EU vs two-state X on the overlapping quarterly sample; report correlations and mean log-level differences; emit two-panel Figure 4.	**ok**	0s

The full machine-readable record is on disk next to this report:

- `verify/verdicts.json` — every claim verdict, with the grader's cited rationale.
- `verify/replication_score.json` — the score and tier breakdown.
- `replication/replication_log.json` — step-by-step execution, with the fixes applied.
- `replication/codebase.diff` — every change made to the provided code.
- `replication/codebase/` — the patched code that was actually run.